CSE2331 Asymptotic Notation Homework

Problem 1. Give the asymptotic complexity of each of the following functions in simplest terms. Your solution should have the form $\Theta(n^\alpha)$ or $\Theta(\log(n)^\beta)$ or $\Theta(n^\alpha \log(n)^\beta)$ or $\Theta(\alpha^n)$ or $\Theta(1)$ where α, β, γ are constants (no justification or proof is needed).

(a) $f(n) = (3n^4 + 8n^2 + 22)^2$ $\Theta(n^8)$

(b) $f(n) = \frac{1}{2}(3n^3 + 6)^2 + 2\left(\frac{3}{7}n + 12\right) + \frac{\sqrt{\frac{6}{7}n+2}}{9}$ $\Theta(n^6)$

(c) $f(n) = \dfrac{6n + 7n^2 + 9}{2n + 6n^{1.3} + n^{1.1}}$ $\Theta(n^2/n^{1.3}) = \Theta(n^{0.7})$

(d) $f(n) = 1222 + \log_3(2384) + \sqrt{e\pi}$ $\Theta(1)$

(e) $f(n) = \frac{22}{7}\log_{12}\left(\frac{n^2}{3} + 2n + 16\right)$ $\Theta(\log(n))$

(f) $f(n) = \underbrace{\left[2(3n + \sqrt{n} + 2)^2 + \frac{16}{3}n\right]}_{\approx n^2} \cdot \underbrace{\log_{12}\left(12n^5 + \frac{66}{5}n^2\right)}_{\approx \log(n)}$ $\Theta(n^2 \log(n))$

(g) $f(n) = 2 \cdot 5^n + 3 \cdot 4^{n+3} + 10$ $\Theta(5^n)$

(h) $f(n) = \underbrace{4 \cdot 5^{2n+3}}_{25^n} + \underbrace{5 \cdot 2^{4n+1}}_{16^n} + \underbrace{3^{3n+0.5}}_{27^n}$ $\Theta(27^n)$

(i) $f(n) = \underbrace{\frac{7}{3} \cdot 6^{2\log_6(n^2+1)}}_{6^{\log_6((n^2+1)^2)} \approx n^4} + \underbrace{9^{\log_2(2n+3)}}_{\approx n^{\log_2(9)}}$ $\Theta(n^4)$

Problem 2. Given the problem above, ORDER the functions in increasing complexity. Your answer should look something like $a < b = c = f < d = g \ldots$

$d < e < c < f < i < b < a < g < h$

Problem 3. Give a function $f(n)$ such that $f(n) \in O(n^3)$ and $f(n) \in \Omega(n^2 \log(n))$, but $f(n) \notin \Theta(n^3)$ and $f(n) \notin \Theta(n^2 \log(n))$.

Many correct answers.

Two common ones:

$f(n) = n^{2.5}$

$f(n) = n^2 \log(n)^2$

Problem 4. Using the definition, prove that the following function

$$f(n) = \log_4(3n^2 + n - 1)\sqrt{n^4 - 8n^2 + 3n + 32}$$

is in $\Theta(n^2 \log(n))$. That is, find $c_1, c_2 > 0$ and $n_0 \in \mathbb{N}$ such that

$$c_1 n^2 \log_4(n) \leq f(n) \leq c_2 n^2 \log_4(n) \qquad \forall n \geq n_0$$

$3n \leq 3n^4$ | $32 \leq 32n^4$

$1 \leq n^3$ | $1 \leq n^4$

$1 \leq n$ | $1 \leq n$

Upper bound:

replace w/ Higher Order Terms.

$$\log_4(3n^2+n-1)\sqrt{n^4 - 8n^2 + 3n + 32} \leq \log_4(3n^2+n^2)\sqrt{n^4 + 3n^4 + 32n^4}$$

can drop − terms

neat idea: $4n^2 \leq n^3$

$4 \leq n$

$$\leq \log_4(4n^2)\cdot\sqrt{36n^4}$$

$$\leq \log_4(n^3)\cdot 6n^2$$

$$\leq 3\log_4(n)\cdot 6n^2$$

Thus, $c_2 = 18$ and $n_0 \geq 4$.

Lower bound:

can drop + terms

$$\log_4(3n^2+n-1)\sqrt{n^4 - 8n^2 + 3n + 32} \geq \log_4(3n^2-n^2)\sqrt{n^4 - \tfrac{1}{2}n^4}$$

$-1 \geq -n^2$

$n^2 \geq 1$

$n \geq 1$

$-8n^4 \geq -\frac{1}{2}n^2$

$16 \leq n^2$

$n \geq 4$

$2n^2 \geq n^2$

$$\geq \log_4(2n^2)\sqrt{n^4/2}$$

$$\geq \log_4(n^2)\cdot\frac{n^2}{\sqrt{2}}$$

$$\geq 2\log_4(n)\cdot\frac{n^2}{\sqrt{2}}$$

$$\geq \sqrt{2}\log_4(n)\,n^2$$

Thus, $c_1 = \sqrt{2}$ and $n_0 \geq 4$.

Choose $n_0 = 4$.

Problem 5. Determine whether each of the following statements is true or false. Explain why the answer is true or false (this does not need to be a formal proof).

(a) **True or false:** If $\lim_{n\to\infty} f(n) = \infty$ and $\lim_{n\to\infty} g(n) = \infty$, then $f(n) \in \Theta(g(n))$.

Counterexample: $f(n) = n^2$, $g(n) = n$

$f(n) \notin \Theta(g(n))$

(b) **True or false:** If $\lim_{n\to\infty} f(n) = 5$ and $\lim_{n\to\infty} g(n) = \infty$, then $f(n) \in O(g(n))$.

By limit theorem!

(c) **True or false:** If $\lim_{n\to\infty} f(n) = \infty$ and $\lim_{n\to\infty} g(n) = 2^{35}$, then $f(n) \in O(g(n))$.

By limit theorem

$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty \Rightarrow f(n) \in \Omega(g(n))$

$f(n) \notin O(g(n))$

(d) **True or false:** If $f(n) \in \Theta(n^2)$ and $g(n) \in \Theta(n)$, then $\frac{f(n)}{g(n)} \in \Theta(n)$.

$$\frac{\Theta(n^2)}{\Theta(n)} = \Theta(n^2/n) = \Theta(n).$$

ANNOYING TO PROVE